Strict global PF_4 for the Riemann kernel

A computer-assisted proof

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https://github.com/falseywinchnet/zeta

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Abstract

Let

$$f(t) = \sum_{n=1}^{\infty} \left(4\pi^2 n^4 e^{9|t|/2} - 6\pi n^2 e^{5|t|/2}\right) e^{-\pi n^2 e^{2|t|}}$$

be the classical Riemann, or de Bruijn–Newman, kernel. We prove that f is strictly PF_4: for every $1 \le r \le 4$ and every strictly increasing real tuples $x_1 < \cdots < x_r$ and $y_1 < \cdots < y_r$, one has

$$\det[f(x_i - y_j)]_{i,j=1}^{r} > 0.$$

The computer-assisted input consists entirely of certified one-variable inequalities, most importantly global positivity of the fully confluent fourth-order invariant $\mathcal{C}_4$. Every subsequent step is exact. A quotient-Wronskian reduction converts arbitrary fourth-order minors into the three-point inequality $\partial_\xi \Psi(\xi; m, r) < 0$. In the global curvature coordinate $y = -(\log f)'$, the relevant numerator becomes

$$\delta\Lambda \int_p^w W(t) \frac{\mathcal{C}_4(t)}{Q(t)^6 \kappa(t)^2}\, \mathrm{d}t.$$

The factors $\delta, \Lambda, Q, \kappa$ are positive, while the CDF crossing kernel W is strictly positive in the interior by a one-crossing density-ratio argument. This proves strict positivity for the full nonconfluent configuration space without a multidimensional interval cover.

Contents

1 Kernel and main result

We use the standard Pólya-frequency terminology of Schoenberg and Karlin [2, 1]. The kernel is

$$f(t) = \sum_{n=1}^{\infty} \left(4\pi^2 n^4 e^{9|t|/2} - 6\pi n^2 e^{5|t|/2}\right) e^{-\pi n^2 e^{2|t|}}, \qquad t \in \mathbb{R}. \tag{1}$$

The theta functional equation gives the smooth even continuation through $t = 0$.

Theorem 1.1 (Strict global PF_4). *For every $1 \le r \le 4$ and every strictly increasing tuples*

$$x_1 < \cdots < x_r, \qquad y_1 < \cdots < y_r,$$

one has

$$\det[f(x_i - y_j)]_{i,j=1}^{r} > 0.$$

Thus the Riemann kernel is a strict PF_4 translation kernel on $\mathbb{R}$.

The proof separates cleanly into two parts. Lemma 2.1 contains the computer-assisted one-variable input. The quotient-Wronskian reduction, curvature-coordinate transformation, transport cancellation, and crossing-kernel argument are exact symbolic and analytic steps.

2 Certified one-variable inputs

Put

$$\ell = \log f, \qquad s = \ell', \qquad q = -s' = -\ell''. \tag{2}$$

Define

$$F_1 = qq'' - (q')^2, \qquad F_2 = q^3 - F_1. \tag{3}$$

Primes in this section denote differentiation in the original variable t.

For $a < b$, set

$$A(a,b) = s(a) - s(b), \qquad M(a,b) = \frac{q(b) - q(a)}{A(a,b)}, \qquad N(a,b) = \frac{q'(b) - q'(a)}{A(a,b)}. \tag{4}$$

For $\xi < m < r$, define the order-three slope functional

$$\Lambda(\xi; m, r) = A(\xi, r) + M(\xi, m) - M(m, r). \tag{5}$$

The fourth-order fully confluent invariant is defined from the logarithmic cumulants $c_j = \ell^{(j)}$, $j = 2, \ldots, 6$. Let

$$\begin{aligned} &m_0 = 1, && m_1 = 0, && m_2 = c_2, && m_3 = c_3, \\ &m_4 = c_4 + 3c_2^2, && m_5 = c_5 + 10c_2c_3, && m_6 = c_6 + 15c_2c_4 + 10c_3^2 + 15c_2^3, \end{aligned}$$

and put

$$\mathcal{C}_4(t) = \det[m_{i+j-2}(t)]_{i,j=1}^{4}. \tag{6}$$

The actual doubly confluent fourth-order kernel minor equals the positive factor $f(t)^4\mathcal{C}_4(t)$ after division by the two positive Vandermonde factors.

Lemma 2.1 (Certified input)**.** *For the kernel f in (1), the following hold on all of $\mathbb{R}$:*

$$q(t) > 0, \qquad F_2(t) > 0, \qquad \mathcal{C}_4(t) > 0. \tag{7}$$

Moreover $\Lambda(\xi; m, r) > 0$ for every $\xi < m < r$.

Proof. The compact interval $[0, 1]$ was certified with directed Arb arithmetic at 192-bit precision. The recorded lower bounds are

$$q \geq 18.726894623395\ldots, \qquad F_2 \geq 3889.215269876\ldots, \qquad \mathcal{C}_4 \geq 28172286.316227\ldots.$$

The q, F_2 certificate uses 7731 second-order Taylor cells; the $\mathcal{C}_4$ certificate uses 8050 cells. Evenness extends these bounds to $[-1, 1]$.

By evenness it is enough to consider $t \geq 1$. Put $x = \pi e^{2t}$. The logarithmic cumulants satisfy

$$c_j = -2^j x + \varepsilon_j, \qquad |\varepsilon_j| \leq \frac{E_j}{x}, \quad j = 2, \ldots, 6, \tag{8}$$

with

$$(E_2, E_3, E_4, E_5, E_6) = (19.8, 176, 2082, 30770, 545900).$$

The corresponding tail estimates give $q > 0$ and $F_2 > 0$. Substitution of (8) into the exact polynomial (56) below yields

$$\frac{\mathcal{C}_4}{x^6} \geq 44392.80919\ldots > 0 \qquad (x \geq \pi e^2),$$

with the exact rational lower-bound polynomial displayed in Appendix A. Evenness then extends the tail conclusions to $t \leq -1$, so the first three inequalities hold globally.

For completeness, the order-three reduction gives the pointwise lower bound

$$\Lambda(\xi; m, r) \geq \int_\xi^r \frac{\min(q(t)^3, F_2(t))}{q(t)^2}\, \mathrm{d}t > 0. \tag{9}$$

This follows by writing the two M terms as q-weighted means of q'/q, then bounding their difference by the endpoint extrema of $(q'/q)' = F_1/q^2$ and using $F_2 = q^3 - F_1$. The symbolic reduction and the directed compact and tail bounds are replayed by the scripts listed in Section 7. ☐

Remark 2.2. The proof below only needs $q > 0$, $\Lambda > 0$, and $\mathcal{C}_4 > 0$. The role of F_2 is to certify $\Lambda > 0$ uniformly, including collision and escape regimes.

3 From arbitrary minors to a three-point derivative

Fix $y_1 < y_2 < y_3 < y_4$ and define translates

$$u_j(t) = f(t - y_j). \tag{10}$$

For $j \geq 2$ put

$$v_j = \left(\frac{u_j}{u_1}\right)', \qquad w_j = \left(\frac{v_j}{v_2}\right)' \quad (j \geq 3). \tag{11}$$

Let $W(g_1, \ldots, g_k)$ denote the Wronskian.

Lemma 3.1 (Quotient identities)**.** *Where the displayed quotients are defined,*

$$W(u_1, u_2, u_3) = u_1^3 W(v_2, v_3), \tag{12}$$

$$W(u_1, u_2, u_3, u_4) = u_1^4 v_2^3 w_3^2 \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{w_4}{w_3}\right). \tag{13}$$

If $U_1 = 1$ and $U_j = u_j/u_1$, then for $k = 3, 4$ and $t_1 < \cdots < t_k$,

$$\det[U_j(t_i)]_{i,j=1}^k = \int_{t_1}^{t_2} \cdots \int_{t_{k-1}}^{t_k} \det[U'_{j+1}(\tau_i)]_{i,j=1}^{k-1}\, \mathrm{d}\tau_1 \cdots \mathrm{d}\tau_{k-1}. \tag{14}$$

The same identity applies after the quotient reduction from v to w.

Proof. The Wronskian identities follow by elementary column operations: divide each column by the first function, differentiate, and repeat. For (14), subtract each row from the row below, beginning at the bottom. Every resulting increment is an integral of U'_j. Multilinearity of the determinant and Fubini's theorem give the formula. Since the intervals $[t_i, t_{i+1}]$ are ordered, every integration point satisfies $\tau_1 < \cdots < \tau_{k-1}$. □

Write

$$p_j = t - y_j, \qquad p_4 < p_3 < p_2 < p_1.$$

Because $q > 0$, s is strictly decreasing and

$$v_j = \frac{u_j}{u_1}\big(s(p_j) - s(p_1)\big) > 0.$$

Let $\mathcal{T}$ denote simultaneous translation of all point arguments. Define

$$\Psi(\xi; m, r) = \Lambda(\xi; m, r) + \mathcal{T} \log \Lambda(\xi; m, r). \tag{15}$$

Proposition 3.2 (Three-point reduction)**.** *For the kernel f, strict global* PF_4 *follows from*

$$\partial_\xi \Psi(\xi; m, r) < 0 \qquad \text{for all } \xi < m < r. \tag{16}$$

In fact, under the already established lower-order positivity, global PF_4 *is equivalent to the weak inequality $\partial_\xi \Psi \le 0$.*

Proof. For $p_3 < p_2 < p_1$, direct differentiation gives

$$\mathcal{T} \log \frac{v_3}{v_2} = \frac{A(p_3, p_2)}{A(p_3, p_1)}\, \Lambda(p_3; p_2, p_1) > 0. \tag{17}$$

Thus $w_j > 0$ for every ordered translate triple, and Lemma 3.1 recovers strict positivity of every order-three collocation minor.

For $j = 3, 4$, let $g_j = \mathcal{T} \log(v_j/v_2)$. Applying the quotient identity one level lower and splitting the M-means at p_2 gives the exact cancellation

$$\mathcal{T} \log \frac{w_4}{w_3} = \Psi(p_4; p_2, p_1) - \Psi(p_3; p_2, p_1). \tag{18}$$

If (16) holds, then $p_4 < p_3$ makes the right-hand side strictly positive. Equation (13) therefore gives positive fourth-order translate Wronskians. Applying (14) twice yields positive fourth-order collocation determinants for arbitrary strictly increasing x- and y-nodes. The converse weak statement follows by taking confluent limits in the x-nodes. □

It remains to prove (16).

4 Curvature coordinate

Since $q > 0$, the function

$$y(t) = -s(t) \tag{19}$$

is a strictly increasing global coordinate, with $\mathrm{d}y/\mathrm{d}t = q(t)$. Define Q by

$$Q(y(t)) = q(t) > 0, \tag{20}$$

and let primes now denote differentiation with respect to y. Put

$$\rho = 1 - Q'', \qquad \kappa = 2 - Q'' = 1 + \rho. \tag{21}$$

Since $Q'' = F_1/q^3$, Lemma 2.1 gives

$$\rho = \frac{F_2}{q^3} > 0, \qquad \kappa = 1 + \frac{F_2}{q^3} > 1. \tag{22}$$

For $\xi < m < r$, set

$$p = y(\xi), \qquad z = y(m), \qquad w = y(r), \qquad L = z - p, \qquad R = w - z. \tag{23}$$

Because $A(a,b) = y(b) - y(a)$ and $M(a,b) = Q[y(a), y(b)]$, equation (5) becomes

$$\Lambda = (w - p) + Q[p,z] - Q[z,w]. \tag{24}$$

Writing $g(y) = Q(y) - y^2$ and using the Peano representation of a second divided difference gives

$$\Lambda = \lambda_L + \lambda_R, \tag{25}$$

$$\lambda_L = \frac{1}{L}\int_p^z (y-p)\kappa(y)\,\mathrm{d}y, \qquad \lambda_R = \frac{1}{R}\int_z^w (w-y)\kappa(y)\,\mathrm{d}y. \tag{26}$$

Also set

$$\delta = -\partial_p \Lambda = \frac{1}{L^2}\int_p^z (z-y)\kappa(y)\,\mathrm{d}y > 0. \tag{27}$$

In y-coordinates, simultaneous translation is the vector field

$$\mathcal{T} = Q(p)\partial_p + Q(z)\partial_z + Q(w)\partial_w. \tag{28}$$

Since $\partial_\xi = Q(p)\partial_p$, equation (27) gives

$$\Lambda_\xi = -Q(p)\delta. \tag{29}$$

Lemma 4.1 (Sign bridge)**.** *Define*

$$\mathcal{N} = \delta\Lambda^2 + \Lambda\big(Q'(p)\delta + \mathcal{T}\delta\big) - \delta\,\mathcal{T}\Lambda. \tag{30}$$

Then

$$\partial_\xi \Psi = -\frac{Q(p)}{\Lambda^2}\mathcal{N}. \tag{31}$$

Equivalently,

$$\mathcal{N} = \delta\Lambda K, \qquad K = \Lambda + Q'(p) + \mathcal{T}\log\delta - \mathcal{T}\log\Lambda. \tag{32}$$

Proof. Differentiate $\Psi = \Lambda + \mathcal{T}\log\Lambda$ with respect to ξ, use (29), commute ∂_ξ with simultaneous translation, and note that

$$\partial_\xi\delta = Q(p)\partial_p\delta, \qquad \partial_\xi Q(p) = Q(p)Q'(p).$$

Collecting over the common denominator Λ^2 gives (31); division by $\delta\Lambda$ gives (32). $\square$

5 The transport identity

Define probability measures on $[p, w]$ by

$$\mathrm{d}\mu(y) = \frac{(z-y)\kappa(y)}{L^2\delta}\,\mathbf{1}_{[p,z]}(y)\,\mathrm{d}y, \tag{33}$$

$$\mathrm{d}\nu(y) = \frac{(y-p)\kappa(y)}{L\Lambda}\,\mathbf{1}_{[p,z]}(y)\,\mathrm{d}y + \frac{(w-y)\kappa(y)}{R\Lambda}\,\mathbf{1}_{[z,w]}(y)\,\mathrm{d}y. \tag{34}$$

The normalizations follow directly from (26) and (27).

Set

$$a(y) = Q(y)(\log\kappa(y))', \qquad b(y) = y - Q'(y), \qquad A_0(y) = 3b(y) - a(y), \tag{35}$$

and

$$D(y) = A_0'(y) = 3(\kappa(y) - 1) - \{Q(y)(\log\kappa(y))'\}'. \tag{36}$$

Lemma 5.1 (Confluent-to-curvature identity)**.** *The invariant* (6) *satisfies*

$$\mathcal{C}_4 = Q^6\kappa^2 D. \tag{37}$$

Consequently $D(y) > 0$ for every $y \in \mathbb{R}$.

Proof. Since $\mathrm{d}/\mathrm{d}t = Q\,\mathrm{d}/\mathrm{d}y$ and $c_2 = -Q$, the cumulants $c_3, \ldots, c_6$ are obtained recursively by applying $Q\,\mathrm{d}/\mathrm{d}y$. Substituting these expressions into the central-moment determinant (6) and factoring gives exactly (37). This is a polynomial identity in $Q, Q', \ldots, Q^{(4)}$ and is independently replayed symbolically. Lemma 2.1, together with $Q > 0$ and $\kappa > 0$, gives $D > 0$. □

For $c < d$, define the chord velocity and chord moment

$$V_{cd}(y) = Q(c) + (y-c)Q[c,d], \tag{38}$$

$$U(c,d) = \int_0^1 ((1-u)Q(c) + uQ(d))\kappa(c + u(d-c))\,\mathrm{d}u. \tag{39}$$

Integration by parts gives the endpoint form

$$U(c,d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d-c} + \frac{(Q(d)-Q(c))^2}{(d-c)^2}. \tag{40}$$

Lemma 5.2 (Exact transport cancellation)**.** *With K as in* (32),

$$K = \mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0]. \tag{41}$$

Equivalently,

$$\frac{\mathcal{N}}{\delta\Lambda} = \mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0]. \tag{42}$$

Proof. Differentiating the triangular integrals along the flow (28) and integrating the κ' terms by parts gives

$$\mathcal{T}\Lambda = U(z,w) - U(p,z), \tag{43}$$

$$\mathcal{T}\delta = \frac{U(p,z) - (Q(z) - Q(p))\delta - Q(p)\kappa(p)}{L}. \tag{44}$$

Substitution in (32) gives the endpoint expression

$$K = \Lambda + Q'(p) - Q[p,z] + \frac{U(p,z) - U(z,w)}{\Lambda} + \frac{U(p,z) - Q(p)\kappa(p)}{L\delta}. \tag{45}$$

To identify the right side as an expectation difference, observe the two exact primitives

$$\kappa A_0 = H', \tag{46}$$
$$H = J', \tag{47}$$

where

$$H(y) = 3y^2 - 3Q(y) - 3yQ'(y) + Q'(y)^2 + Q(y)Q''(y), \tag{48}$$
$$J(y) = y^3 - 3yQ(y) + Q(y)Q'(y). \tag{49}$$

Let $I_L = \int_p^z H$ and $I_R = \int_z^w H$. One integration by parts in the linear weights of (33)–(34) yields

$$\mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0] = \frac{-I_L/L + I_R/R}{\Lambda} + \frac{LH(p) - I_L}{L^2\delta}. \tag{50}$$

Using $H = J'$, inserting the endpoint values of J, and simplifying with (40) transforms (50) exactly into (45). This proves (41) and (42) for arbitrary smooth Q; no estimate is used. □

6 Positive crossing kernel and completion

Let F_μ, F_ν be the distribution functions of (33)–(34), and define

$$W(t) = F_\mu(t) - F_\nu(t). \tag{51}$$

Lemma 6.1 (Strict stochastic ordering)**.** *One has*

$$W(t) > 0 \quad (p < t < w), \qquad W(p) = W(w) = 0. \tag{52}$$

Proof. On $[z,w]$, $F_\mu = 1$, hence $W = 1 - F_\nu > 0$ before the right endpoint. On (p,z), the density ratio is

$$\frac{\mathrm{d}\mu/\,\mathrm{d}y}{\mathrm{d}\nu/\,\mathrm{d}y} = \frac{\Lambda(z-y)}{L\delta(y-p)}. \tag{53}$$

It decreases strictly from $+\infty$ to 0. Therefore $W' = \mu' - \nu'$ changes sign exactly once, from positive to negative. Since $W(p) = 0$ and

$$W(z) = 1 - F_\nu(z) = \nu([z,w]) > 0,$$

the function cannot return to zero on $(p,z]$. This proves the claim. □

Integration by parts for distribution functions, using $A_0' = D$, gives

$$\mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0] = \int_p^w W(t)D(t)\ \mathrm{d}t. \tag{54}$$

Combining Lemmas 5.1, 5.2, and 6.1, we obtain the central identity

$$\boxed{\mathcal{N} = \delta\Lambda \int_p^w W(t)\frac{\mathcal{C}_4(t)}{Q(t)^6\kappa(t)^2}\ \mathrm{d}t > 0.} \tag{55}$$

Every factor is positive in the interior, and the interval has positive length. Lemma 4.1 therefore yields

$$\partial_\xi \Psi(\xi; m, r) < 0 \qquad (\xi < m < r).$$

Proposition 3.2 proves Theorem 1.1.

Remark 6.2. The theorem is a finite-order total-positivity statement. No implication for the Riemann Hypothesis is used or claimed here.

7 Reproducibility and focused audit

The source repository is `https://github.com/falseywinchnet/zeta`. This draft was prepared from commit

`7353231a516838cd439dc0f5ab44948f6fcab566`.

The principal proof records are:

Purpose	Repository file
Global q, F_2 and PF_3 input	`sources/pf3-global-certificate.md`
Global confluent $\mathcal{C}_4 > 0$	`sources/c4-confluent-certificate.md`
Wronskian and three-point reduction	`sources/pf4-wronskian-reduction-certificate.md`
Transport identity and final bridge	`sources/pf4-transport-kernel-certificate.md`
Exact symbolic replay	`scripts/verify_pf4_transport_kernel.py`
$\mathcal{C}_4$ replay	`scripts/verify_c4_confluent.py`
PF_3 replay	`scripts/verify_pf3_reduction.py`

The certificate environment records `sympy==1.14.0`, `mpmath==1.3.0`, and `python-flint==0.9.0`. The transport replay is exact symbolic algebra with independent 70-digit kernel checks. The compact one-variable inequalities use directed Arb balls at 192-bit precision.

For a rapid independent review, the highest-value checks are:

1. Verify the kernel normalization in (1) and the sign conventions in the confluent determinant.
2. Re-derive (18) from the quotient identities and confirm that $\partial_\xi \Psi < 0$ produces the positive orientation in (13).
3. Verify the algebraic identity (37) directly from (6).
4. Check the two primitives (46)–(49) and the endpoint reduction from (50) to (45).
5. Independently replay the directed compact certificates and the explicit tail polynomial in Appendix A.

A Exact fourth-order formulas and certificate data

Expanding the central-moment determinant (6) gives

$$\begin{aligned}\mathcal{C}_4 = 12c_2^6 + 24c_2^4c_4 - 24c_2^3c_3^2 + 2c_2^3c_6 - 12c_2^2c_3c_5 + 7c_2^2c_4^2 \\ + 12c_2c_3^2c_4 + c_2c_4c_6 - c_2c_5^2 - 9c_3^4 - c_3^2c_6 + 2c_3c_4c_5 - c_4^3.\end{aligned} \tag{56}$$

Under (8), exact rational coefficient collection gives the lower-bound polynomial

$$\begin{aligned}P(x) = 49152x^6 - \frac{7299072}{5}x^4 - \frac{88018944}{5}x^3 - \frac{756929024}{5}x^2 - \frac{9307871232}{25}x \\ - \frac{90509073984}{25} - \frac{1807239327472}{125x} - \frac{3528006311808}{125x^2} - \frac{62513985514124}{625x^3} \\ - \frac{220885105972212}{3125x^4} - \frac{8406623961648}{625x^5} - \frac{11297761792812}{15625x^6}.\end{aligned}$$

For $x \geq \pi e^2 > 1157/50$,

$$\mathcal{C}_4(t) \geq P(x) > 0,$$

and the normalized monotone margin is

$$\frac{P(x)}{x^6} \geq \frac{2554541341882650566054082955456988112363 52}{575440344669941201115023026380055 5601} = 44392.80919\ldots.$$

The directed compact certificate records

$$\mathcal{C}_4(t) \geq 28172286.3162273782344522118100245449538 1 \qquad (0 \leq t \leq 1).$$

References

[1] S. Karlin, *Total Positivity, Vol. I*, Stanford University Press, 1968.

[2] I. J. Schoenberg, *On Pólya frequency functions. I. The totally positive functions and their Laplace transforms*, J. Analyse Math. 1 (1951), 331–374.

[3] falseywinchnet, *zeta: MIND research repository*, `https://github.com/falseywinchnet/zeta`, commit `7353231a516838cd439dc0f5ab44948f6fcab566`, accessed 15 July 2026.